

Particle Physics Homework 1

1) Beirut Explosion

On 4th August 2020, an explosion occurred at Beirut, the capital city of Lebanon. We can use the dimensional analysis technique developed by Geoffrey Taylor (1945) to estimate the total energy output of the bomb.

The analysis can be done by proposing that there are only 4 parameters in the physics of explosion:

E = total energy output, t = time

R = blast radius, ρ = air density.

We can work out the equation as

$$R = E^{\alpha} \rho^{\beta} t^{\gamma}$$

1.1) Work out the power α , β , and γ as a number using dimensional analysis (no need for natural unit)

1.2) The Beirut's explosion was captured by video and one of the frame has been analysed such that

At $t = 0.233$ s, the blast radius is 179.97 m.

The air density at that time is 1.23 kg/m^3 and

1 million kilogram of TNT (kt) = $4.18 \times 10^{12} \text{ J}$.

Find the energy output in kt.

2). Natural Unit

- 2.1). Find $\frac{1}{m_e}$ in metre (This is Compton wavelength of e^-)
- 2.2). Find $\frac{1}{m_p}$ in metre (This is the characteristic size of a nucleus)
- 2.3). Find the probing scale (in metre) of the LHC at 14 TeV
- 2.4). Express Newtons constant G_N in eV^n and s^m
- 2.5). Compare the relative strength of F_{em} & $F_{gravity}$ between e^+ & e^-
- 2.6). Approximate the effective ranges of the weak force, the strong force between nucleus and the Higgs force.
- 2.7). Estimate the number of protons in the Atlantic ocean.
(You are allowed to use google for relevant informations)

3). Vector Calculus

Identities in Vector Calculus are a lot easier to prove in the index form. Let's learn how to prove them in this exercise.

The notation of vector in the index form is given by:

$$\vec{A} \cdot \vec{B} = A_i B_i \quad (\text{inner product})$$

$$(\vec{A} \otimes \vec{B})_{ij} = A_i B_j \quad (\text{tensor product})$$

$$(\vec{A} \vec{x})_i = A_{ij} x_j \quad (\text{matrix multiplication})$$

$$\text{tr}(A) = A_{ii} \quad (\text{trace})$$

$$(A^T)_{ij} = A_{ji} \quad (\text{transpose})$$

Another useful symbol is the Levi-Civita symbol, ϵ_{ijk} , defined as

i) $\epsilon_{ijk} = 0$ if any two indices are equal ($\epsilon_{112} = \epsilon_{131} = \epsilon_{332} = 0$)

ii) $\epsilon_{ijk} = -\epsilon_{jik} = -\epsilon_{kji} = -\epsilon_{ikj}$

(It picks up a minus sign for any odd number of permutation

e.g. $\epsilon_{123} \xrightarrow{1 \text{ time}} -\epsilon_{213} \xrightarrow{2 \text{ times}} \epsilon_{231}$

iii) $\epsilon_{123} = 1$

This symbol is useful for writing the cross product

Consider 2 vectors: $\vec{A} = A_i \hat{e}_i$, $\vec{B} = B_i \hat{e}_i$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = \hat{e}_1(A_2 B_3 - A_3 B_2) + \hat{e}_2(A_3 B_1 - A_1 B_3) + \hat{e}_3(A_1 B_2 - A_2 B_1)$$

$\uparrow \epsilon_{123} \quad \uparrow \epsilon_{132} \quad \uparrow \epsilon_{231} \quad \uparrow \epsilon_{213} \quad \uparrow \epsilon_{312} \quad \uparrow \epsilon_{321}$

$$= \epsilon_{ijk} \hat{e}_i A_j B_k \quad \therefore (\vec{A} \times \vec{B})_i = \epsilon_{ijk} A_j B_k$$

The product of 2 Levi-Civita symbols is written as

$$\epsilon_{ijk} \epsilon_{lmn} = \delta_{il} \delta_{jm} \delta_{kn} + \delta_{im} \delta_{jn} \delta_{kl} + \delta_{in} \delta_{jl} \delta_{km} - \delta_{im} \delta_{jl} \delta_{kn} - \delta_{il} \delta_{jn} \delta_{km} - \delta_{in} \delta_{jm} \delta_{kl}$$

Now prove the following identities:

3.1) $\epsilon_{ijk} \epsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}$

3.2) $\epsilon_{ijk} \epsilon_{ijn} = 2 \delta_{kn}$

3.3) $\epsilon_{ijk} \epsilon_{ijk} = 6$

3.4) $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$

$$3.5) \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \vec{\nabla}^2 \vec{A}$$

$$3.6) \vec{\nabla} \times (\vec{\nabla} f) = 0$$

$$3.7) \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

$$3.8) \vec{\nabla} \times (\vec{A} \times \vec{B}) = \vec{A}(\vec{\nabla} \cdot \vec{B}) - (\vec{\nabla} \cdot \vec{A})\vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} - (\vec{A} \cdot \vec{\nabla})\vec{B}$$

$$3.9) \vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{A} \cdot (\vec{\nabla} \times \vec{B}) - \vec{B} \cdot (\vec{\nabla} \times \vec{A})$$

$$3.10) \vec{\nabla}(\vec{A} \cdot \vec{B}) = \vec{A} \times (\vec{\nabla} \times \vec{B}) - (\vec{\nabla} \times \vec{A}) \times \vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} + (\vec{A} \cdot \vec{\nabla})\vec{B}$$

$$3.11) \vec{A} \times (\vec{B} \times \vec{C}) + \vec{C} \times (\vec{A} \times \vec{B}) + \vec{B} \times (\vec{C} \times \vec{A}) = 0$$

$$3.12) \vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$3.13) (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{B} \cdot \vec{C})(\vec{A} \cdot \vec{D})$$

$$3.14) (\vec{A} \cdot (\vec{B} \times \vec{C}))\vec{D} = (\vec{A} \cdot \vec{D})(\vec{B} \times \vec{C}) + (\vec{B} \cdot \vec{D})(\vec{C} \times \vec{A}) + (\vec{C} \cdot \vec{D})(\vec{A} \times \vec{B})$$

$$3.15) (\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = (\vec{A} \cdot (\vec{B} \times \vec{D}))\vec{C} - (\vec{A} \cdot (\vec{B} \times \vec{C}))\vec{D}$$